API Endpoint Plan

Student-Teacher Course Management System | PROG6212 Teaching Example

The following endpoints support authentication, student and teacher profiles, courses, course categories, enrolments and results.

Role Required: None = public access;

Any = logged-in Student or Teacher.

All endpoints use the appropriate HTTP status codes.

A. Authentication

HTTP Method	Route	Description	Role Required	Request Body (if any)	Expected Response
POST	/api/auth/register	Register a Student or Teacher; hash password and create the correct linked profile.	None	Common: { email, password, role, firstName, lastName }; Student: { studentNumber }; Teacher: { employeeNumber }	201 Created – account/profile; 400 invalid input/role; 409 duplicate email or number
POST	/api/auth/login	Validate credentials and issue the existing cookie-based authentication ticket.	None	{ email, password }	200 OK – login confirmation and cookie; 401 invalid credentials
POST	/api/auth/logout	Sign/log out and clear the authentication cookie.	Any	None	200 OK – logged out; 401 if authentication is required

B. User Profile

HTTP Method	Route	Description	Role Required	Request Body (if any)	Expected Response
GET	/api/auth/profile	View the logged-in Student or Teacher profile and account details.	Any	None	200 OK – own profile; 401 Unauthorized
PUT	/api/auth/profile	Update the logged-in user's own profile details.	Any	{firstName, lastName }	200 OK – updated profile; 400 invalid inputs; 401 Unauthorized

C. Courses

HTTP Method	Route	Description	Role Required	Request Body (if any)	Expected Response
GET	/api/courses	Get all courses available to browse.	None	None	200 OK – course list
GET	/api/courses/{id}	Get details of a particular course, including category and Teacher.	None	None	200 OK – course; 404 Not Found
GET	/api/courses/my	Get courses created by the logged-in Teacher for their dashboard.	Teacher	None	200 OK – Teacher course list; 403 Forbidden
POST	/api/courses	Create a course assigned to the logged-in Teacher and selected category.	Teacher	{courseCategoryId, code, title, description }	201 Created – course; 400 invalid data/category; 409 duplicate code
POST	/api/courses/{id}/image	Upload or replace the image for a course owned by the logged-in Teacher. The API stores the image in Azure Blob Storage and updates CourseImageUrl.	Teacher	multipart/form-data containing an image file	200 OK - image URL and updated course; 400 - invalid file; 403 - not course owner; 404 - course not found
PUT	/api/courses/{id}	Update a course owned by the logged-in Teacher.	Teacher	{courseCategoryId, code, title, description }	200 OK – updated course; 400 invalid data; 403 not

					owner; 404 missing; 409 duplicate code
DELETE	/api/courses/{id}	Delete a course owned by the Teacher, if no enrolments depend on it.	Teacher	None	204 No Content; 403 not owner; 404 missing; 409 dependent enrolments

D. Course Categories

HTTP Method	Route	Description	Role Required	Request Body (if any)	Expected Response
GET	/api/categories	View available course categories shared across courses.	None	None	200 OK – category list
POST	/api/categories	Create a global course category.	Teacher	{name, description }	201 Created – category; 400 invalid data; 409 duplicate name
PUT	/api/categories/{id}	Update an existing course category.	Teacher	{name, description }	200 OK – updated category; 400 invalid data; 404 missing; 409 duplicate name
DELETE	/api/categories/{id}	Delete a category only if it is not used by courses.	Teacher	None	204 No Content; 404 missing; 409 category in use

E. Course Enrolments

HTTP Method	Route	Description	Role Required	Request Body (if any)	Expected Response
POST	/api/courses/{id}/enrol	Enrol the logged-in Student in the selected course; initial status Pending.	Student	None	201 Created – enrolment; 404 course missing; 409 already enrolled
GET	/api/enrolments/my	Get all enrolments for the	Student	None	200 OK – own

		logged-in Student, with course and status.			enrolments
GET	/api/enrolments/{id}	Get one enrolment belonging to the logged-in Student.	Student	None	200 OK – enrolment; 403 not owner; 404 missing
GET	/api/courses/{id}/enrolments	Get enrolments for a course owned by the logged-in Teacher.	Teacher	None	200 OK – enrolments and Student details; 403 not owner; 404 course missing
PUT	/api/enrolments/{id}/status	Update enrolment status for a course owned by the logged-in Teacher.	Teacher	{status } (Pending, Confirmed or Cancelled)	200 OK – updated enrolment; 400 invalid transition; 403 not owner; 404 missing
DELETE	/api/enrolments/{id}	Withdraw the logged-in Student's own enrolment before a result exists.	Student	None	204 No Content – withdrawn; 403 not owner; 404 missing; 409 result already recorded

F. Course Results

HTTP Method	Route	Description	Role Required	Request Body (if any)	Expected Response
POST	/api/enrolments/{id}/result	Teacher records a final mark for a confirmed enrolment in their course; API derives Pass/Fail.	Teacher	{ finalMark }	201 Created – result; 400 invalid mark/status; 403 not owner; 404 missing; 409 result exists
GET	/api/results/my	Get the logged-in Student's results with course, final mark and outcome.	Student	None	200 OK – own results
GET	/api/courses/{id}/results	Get recorded results for a course, restricted to its Teacher.	Teacher	None	200 OK – course results; 403 not owner; 404 missing
PUT	/api/results/{id}	Teacher corrects a result for their course; API recalculates	Teacher	{ finalMark }	200 OK – updated result; 400 invalid mark; 403 not

	Pass/Fail.	owner; 404 missing